Deep Learning-Based Medication Recommendation System for Primary Care Decision Support

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Surveyed Studies (1–7)

Study / Year	Arch.	Safety	Limitation
DP-CRNN (2021) [19]	CNN+RNN	No	Not drug-centric
Deep Patient (2016) [13]	Autoenc.	No	Static reps
Doctor AI (2016) [5]	RNN	No	Low interpretability
RETAIN (2016) [7]	Attn-RNN	No	Not drug-aware
GAMENet (2019) [15]	GNN+Mem	Yes	Dense EHR needed
SafeDrug (2021) [12]	Mol-GNN	Yes	Compute heavy
GRAM (2017) [6]	Ontology GAT	Partial	Ontology dependent

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Surveyed Studies (8–14)

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MoleRec (2022–23) [2]	GNN	Yes	Complex training
ClinicalBERT (2019) [3]	Transformer	No	Pretrain cost
MemoryNet (2019) [16]	Memory NN	No	Needs history
HAN (2016) [17]	Hier. Attn	No	Complex
GRU-D (2018) [4]	GRU-Decay	No	Vitals focused
Polypharmacy GNN (2018) [20]	GNN	Yes	Graph sensitive

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RL Treatments (2018) [10]	RL	Partial	Reward design
LIME / SHAP [14]	Explainers	No	Approximate
KG Embeddings [1]	KG models	Yes	KG curation
Clinical Transformers (2022) [9]	Transformer	Partial	Resource heavy

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System Overview

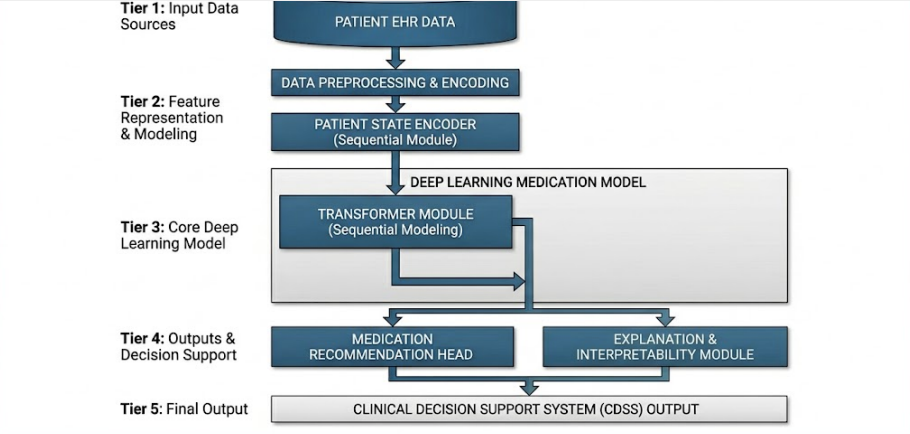
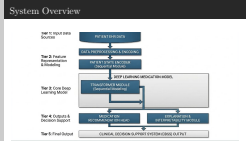


Figure: Four-phase medication recommendation pipeline.



Walk through the figure – four phases: preprocessing, representation learning, recommendation + safety, interpretability.

Preprocessing & Patient Representation

Input: Free-text symptoms + age, sex, comorbidities

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Algorithm

Algorithm 1 DL Medication Recommendation

Require: S (symptom text), D (structured data), G (DDI graph)
Ensure: M_{safe}, explanation report E

1: Normalize S: spell correction + NER $\rightarrow$ ontology mapping
2: $E_s \leftarrow \text{ClinicalDistilBERT}(S)$; $\quad E_d \leftarrow \text{MLP}(D)$
3: $P_v \leftarrow \text{Concatenate}(E_s, E_d)$
4: Sigmoid classifier $\rightarrow$ top-k candidates
5: DDI safety filter over $G \rightarrow M_{safe}$
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Condensed from the full algorithm – hit the key steps. Good slide to have open during Q&A.

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Training Convergence

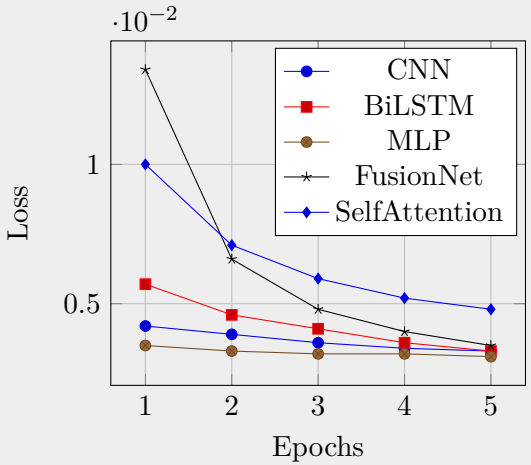
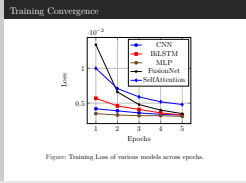


Figure: Training Loss of various models across epochs.



MLP lowest absolute loss but simpler model. FusionNet starts high, converges fast once text encoder stabilizes – key sign the dual-branch design works.

Model Performance — Micro and Macro Metrics

Model	Jaccard	Precision$_\mu$	Recall$_\mu$	F1$_\mu$
MLP	0.6887	0.8853	0.7561	0.8156
CNN	0.7657	0.9031	0.8342	0.8673
BiLSTM	0.7796	0.9218	0.8348	0.8762
SelfAttention	0.7684	0.9022	0.8382	0.8691
FusionNet	0.7869	0.9252	0.8404	0.8808
Model	**Hamming**	**Recall$_M$**	**F1$_M$**	**Precision$_M$**
MLP	0.0015	0.0675	0.0721	0.0781
CNN	0.0011	0.0762	0.0798	0.0858
BiLSTM	0.0010	0.0777	0.0821	0.0899
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Model Performance — Error and AUC

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CNN	0.5603	0.9968	0.8743
BiLSTM	0.6184	0.9957	0.8743
SelfAttention	0.5675	0.9970	0.8764
FusionNet	0.6455	0.9964	0.8753
Model	**Precision@K**	**Recall@K**	**F1@K**
MLP	0.6855	0.6329	0.5825
CNN	0.7101	0.6563	0.6042
BiLSTM	0.7076	0.6555	0.6028
SelfAttention	0.7114	0.6573	0.6052
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The subset accuracy point is the most clinically meaningful – drive this home. Macro-F1 is a known limitation of the synthetic data, not a flaw in the model.

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- High ROC-AUC across all models: the symptom–medication signal is learnable even by shallow models

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Future Work

- Evaluate on real-world clinical corpora
- Runtime DDI filtering with quantifiable rate reduction
- Class-imbalance strategies for rare drug prediction
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Good slide to end on – shows the work is a starting point, not a dead end. The LM swap is a quick win worth highlighting.

Authors' Contributions

Table: Dataset Generation Contribution

Author	Dataset / Task	Model
Arnav Sahu	Disease–Drug	MLP
Pratyush Bhagat	Symptom–Disease	Self Attention
Nakul Bhat	Drug–Drug Interaction	BiLSTM
Hrithiq Gupta	Dataset Generation	FusionNet, CNN

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